

SANKALP JETTY

Senior Data Engineer | Data Architecture & Real-Time Systems Texas, USA | (469) 4753-469 |
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PROFESSIONAL SUMMARY

Highly certified and accomplished Data Engineer with 5+ years of progressive experience designing, deploying, and optimizing robust, high-volume data architectures (AWS/hybrid). Expertise in **real-time streaming, high-scale ETL/ELT, and enterprise data governance**. Key achievements include **\$180K annual cost savings, 30% performance gains** via Databricks Photon migration, **sub-5 second latency** with Kafka/Snowflake, and **65% reduction** in reporting errors using Great Expectations. Seeking a senior role leveraging deep experience in Spark, Kafka, Snowflake and dbt.

PROFESSIONAL EXPERIENCE

Capital One | Data Engineer | Texas, USA Aug 2023 – Present

- Established **Data Mesh architecture principles** for domain ownership, enabling independent deployment of data products and reducing inter-team dependency by 35%.
- Designed and implemented **Schema Evolution** protocols using Avro serialization for Kafka topics, ensuring backward compatibility across 10+ consumer applications.
- Architected high-throughput **Airflow** orchestration for ETL/ELT workflows using **Apache Spark** to ingest and transform **25+ TB/day** of critical data (fraud, credit, customer analytics) across AWS.
- Optimized data marts and advanced models (**Dimensional, Data Vault**) in **Snowflake, Redshift, and SQL Server**, supporting self-service analytics for 150+ analysts.
- Partnered with Data Science teams to streamline feature engineering and integrate production ML models into automated MLOps pipelines.
- Mentored 8 junior engineers on **dbt modeling, Spark optimization, and secure CI/CD pipelines**, increasing team velocity by 40%.
- Implemented end-to-end observability (OpenTelemetry, Prometheus, Grafana), cutting Mean Time to Recovery (MTTR) from 2 hours to under **30 minutes**.

Freshworks | Data Engineer | Hyderabad, India Jan 2020 – Jul 2022

- Architected and enforced **Role-Based Access Control (RBAC)** policies in Snowflake and Redshift, minimizing data exposure risks and ensuring strict compliance with **HIPAA and SOC 2**.
- Led the migration of **50+ TB** of legacy healthcare data from on-prem Hadoop HDFS to **Snowflake**, securing highly available data storage using **AWS S3 data lakes**.
- Designed streaming infrastructure using **Apache Kafka** to ingest **10+ TB/day** of healthcare data (EHRs, IoT medical devices), enabling real-time operational analytics.
- Developed scalable ETL pipelines using **AWS Glue** and **Spark SQL**, **reducing data transformation time by 45%** and ensuring data governance.
- Spearheaded performance tuning on the Redshift cluster by implementing columnar storage and sort keys, reducing the average execution time of critical BI queries by over 400ms.
- Delivered high-impact business solutions by designing interactive **Power BI dashboards** visualizing key clinical KPIs, resulting in a **25% improvement in operational efficiency**.

TECHNICAL SKILLS

Platforms & Processing: Snowflake, Databricks, AWS (S3, Redshift, Glue, EKS, Lambda, Sagemaker), Azure, GCP, Google BigQuery, **Azure Synapse**, **Apache Spark**, **Kafka** (Confluent Cloud), Apache Hadoop (HDFS, Hive, HBase), Apache Beam, ETL/ELT, Streaming Data, Big Data Processing.

Data Movement & Databases: Fivetran, Stitch, Matillion, **Apache NiFi**, StreamSets, **Teradata**, **DB2**, PostgreSQL, MySQL, SQL Server, Oracle, MongoDB. **SSIS**, **SSRS**, Informatica, Talend, Azure Data Factory (ADF).

Visualization & Governance: Tableau, Power BI, QuickSight, **Looker**, **dbt**, Great Expectations, Collibra, Data Vault, Dimensional Modeling, Schema Evolution, Data Governance.

Engineering & MLOps: **Python** (Pandas, NumPy), **SQL** (Advanced), **Airflow**, Jenkins, Docker, Kubernetes, Terraform, **Prometheus**, **OpenTelemetry**, **Grafana**, **ELK Stack** (Elasticsearch, Logstash, Kibana), MLOps, CI/CD, Agile.

EDUCATION & CERTIFICATIONS

Master of Science in Management Science | University of Texas at Dallas, Texas, USA

Bachelor of Technology in Chemical Engineering | National Institute of Technology Warangal (NITW), India

Certifications: AWS Certified Data Engineer - Associate, Snowflake SnowPro Core, Databricks Data Engineer Associate, AWS Certified Cloud Practitioner